main.py

Visualize

Run

```python
import pandas as pd

data = {
    "Name": ["Arun", "Bala", "Charan", "Divya"],
    "Python": [80, 65, 90, 75],
    "Java": [85, 70, 88, 80],
    "DBMS": [75, 60, 92, 78]
}

df = pd.DataFrame(data)

df["Total"] = df["Python"] + df["Java"] + df["DBMS"]
df["Average"] = df["Total"] / 3

print(df[["Name", "Total", "Average"]].to_string(
    index=False,
    formatters={"Average": "{:.2f}".format}
))

print("\nStudents with Average > 75:")
result = df[df["Average"] > 75]

print(result[["Name", "Average"]].to_string(
```

Output

```
  Name  Total Average
  Arun    240   80.00
  Bala    195   65.00
Charan    270   90.00
 Divya    233   77.67

Students with Average > 75:
  Name Average
  Arun   80.00
Charan   90.00
 Divya   77.67

=== Code Execution Successful ===
```

main.py Visualize Run

```python
import pandas as pd

data = {
    "Product": ["Laptop", "Mouse", "Keyboard", "Mobile", "Tablet
    "Category": ["Electronics", "Accessories", "Accessories",
                 "Electronics", "Electronics"],
    "Quantity": [5, 20, 10, 8, 6],
    "Price": [50000, 500, 1000, 20000, 15000]
}

df = pd.DataFrame(data)

df["Total_Sales"] = df["Quantity"] * df["Price"]

print("Product Sales:")
print(df[["Product", "Total_Sales"]].to_string(index=False))

category_sales = df.groupby("Category")["Total_Sales"].sum()

print("\nCategory Total Sales:")
print(category_sales.to_string())
```

Output

```
Product Sales:
 Product  Total_Sales
  Laptop       250000
   Mouse        10000
Keyboard        10000
  Mobile       160000
  Tablet        90000

Category Total Sales:
Category
Accessories     20000
Electronics    500000

=== Code Execution Successful ===
```

main.py

Visualize

Run

```python
import pandas as pd
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya"],
    "Total_Days": [50, 50, 50, 50],
    "Present_Days": [45, 38, 48, 35]
}

df = pd.DataFrame(data)
df["Attendance %"] = (
    df["Present_Days"] / df["Total_Days"]
) * 100
print(df[["Name", "Attendance %"]].to_string(index=False))
print("\nStudents below 80%:")
result = df[df["Attendance %"] < 80]
print(result[["Name", "Attendance %"]].to_string(index=False))
```

Output

```
  Name  Attendance %
  Arun          90.0
  Bala          76.0
Charan          96.0
 Divya          70.0

Students below 80%:
 Name  Attendance %
 Bala          76.0
Divya          70.0

=== Code Execution Successful ===
```

main.py

Visualize

Run

```python
import pandas as pd

data = {
    "Name": ["Arun", "Bala", "Charan", "Divya"],
    "Total_Days": [50, 50, 50, 50],
    "Present_Days": [45, 38, 48, 35]
}

df = pd.DataFrame(data)

df["Attendance %"] = (
    df["Present_Days"] / df["Total_Days"]
) * 100

print(df[["Name", "Attendance %"]].to_string(index=False))

print("\nStudents below 80%:")

result = df[df["Attendance %"] < 80]

print(result[["Name", "Attendance %"]].to_string(index=False))
```

Output

```
  Name  Attendance %
  Arun          90.0
  Bala          76.0
Charan          96.0
 Divya          70.0

Students below 80%:
 Name  Attendance %
 Bala          76.0
Divya          70.0


=== Code Execution Successful
```

main.py

Visualize

Run

```python
import pandas as pd

data = {
    "ID": [101, 102, 103, 104, 105],
    "Name": ["Ravi", "Kumar", "Anu", "Priya", "Manoj"],
    "Department": ["IT", "HR", "IT", "HR", "Sales"],
    "Salary": [50000, 45000, 65000, 55000, 60000]
}

df = pd.DataFrame(data)

result = df.loc[df.groupby("Department")["Salary"].idxmax()]

result = result.sort_values("Department")

print(result[["Department", "Name", "Salary"]].to_string(index=False))
```

Output

```
Department  Name  Salary
        HR Priya   55000
        IT   Anu   65000
     Sales Manoj   60000

=== Code Execution Successful ===
```